INDICATION FOR PROCEDURE: A 63-year-old male with long-standing hypertension, dyslipidemia, and recurrent Canadian Cardiovascular Society class III angina pectoris despite guideline-directed medical therapy. Positive nuclear myocardial perfusion SPECT demonstrating high-risk reversible anterior and apical ischemia.

PROCEDURE PERFORMED:

1. Selective coronary arteriography via right radial artery access using a 6-French Glidesheath.

2. Left ventriculography with resting hemodynamic pressure measurements.

3. Percutaneous transluminal coronary angioplasty and deployment of a 3.0 x 24 mm Promus Premier drug-eluting stent to the proximal-to-mid left anterior descending (LAD) coronary artery.

DESCRIPTION OF PROCEDURE: The right radial artery was cannulated under local 1% lidocaine anesthesia utilizing modified Seldinger technique. A 6-French radial sheath was inserted over a 0.021-inch wire. An intra-arterial cocktail consisting of 5,000 units unfractionated heparin, 2.5 mg verapamil, and 200 mcg nitroglycerin was administered. Hemodynamic measurements revealed aortic root pressure of 134/78 mmHg, left ventricular end-diastolic pressure (LVEDP) elevated at 22 mmHg. Selective angiography of the left coronary system in standard orthogonal caudal and cranial projections demonstrated normal left main coronary artery bifurcating into LAD and circumflex. The LAD revealed a severe, concentric, calcified 85% stenosis in the proximal-to-mid segment involving a major diagonal branch takeoff (Medina 1,1,0). Left circumflex artery demonstrated mild luminal irregularities without hemodynamically flow-limiting lesions. Right coronary artery was dominant with smooth contours.

The decision was made to proceed with ad-hoc PCI. A 6-French EBU 3.5 guide catheter was seated in the left main ostium. A 0.014-inch Runthrough guide wire successfully traversed the LAD lesion into the distal vessel. Predilation was performed using a 2.5 x 15 mm semi-compliant balloon inflated to 12 atmospheres. A 3.0 x 24 mm drug-eluting stent was navigated across the stenotic segment and deployed at 14 atmospheres for 30 seconds. Post-dilation with a 3.25 x 12 mm non-compliant NC Trek balloon inflated to 18 atmospheres demonstrated excellent stent apposition without edge dissection, residual stenosis zero percent, and restoration of TIMI grade 3 distal perfusion. Right radial sheath was removed with a TR Band compression device applied. The patient tolerated the procedure without arrhythmia or complications.